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Report on

Predictive Modeling and Fairness in Health Insurance Pricing

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Abstract

Machine learning models are widely used to predict medical costs and guide insurance pricing. However, in high-stakes applications, accuracy alone is not sufficient. We must also ensure that model predictions do not unfairly disadvantage specific groups.

Using a synthetic but realistic U.S. health insurance dataset (Kaggle source; see Section 2), we study smoking status as a sensitive attribute. We answer two questions: (1) Do smokers have higher annual medical charges in the data? (2) Does a predictive model reproduce these cost differences without adding additional unfair bias?

We build a complete, reproducible pipeline: data cleaning, encoding, train–test split, model training, model comparison, and a multi-view fairness evaluation. Fairness is assessed (i) in regression space using residual statistics and mispricing rates, (ii) with bootstrap confidence intervals for group differences, and (iii) with group fairness metrics from the course (Demographic Parity, Equal Opportunity, Equalized Odds) after converting charges into a high-cost classification task. Finally, we validate our fairness diagnostics using a stress test in which we inject an artificial surcharge (+5000 $) for smokers.

Our results show large *true* charge differences between smokers and non-smokers in the dataset, while the baseline predictive model has nearly identical error patterns for both groups. The bootstrap confidence intervals for group differences in error metrics include zero, and the stress test confirms that our fairness pipeline detects discrimination when it is present.

1 Introduction

Insurance pricing often relies on predicted costs. A predictive model is useful only if it performs well on new applicants. However, in insurance and healthcare, accuracy alone is not sufficient. Models must also be evaluated for fairness and responsible use, since their predictions can directly affect prices and access to services.

In this project, we focus on **smoking status**. Smoking is a significant risk factor for health outcomes, so some cost differences may be medically justified. At the same time, fairness concerns arise if a model introduces additional penalties that are not justified by observed data patterns, or if it produces systematically larger errors for one group.

1.1 Scientific questions

We study two questions:

Q1. Cost difference (descriptive): Do smokers systematically incur higher annual insurance charges than non-smokers in the dataset?

Q2. Algorithmic fairness (predictive): Does a predictive model reproduce cost differences *without introducing additional bias*, meaning similar error patterns for smokers and non-smokers?

1.2 Ethical considerations (what "fair" means here)

This project does *not* claim that the use of smoking status in insurance pricing is ethically or legally acceptable in all contexts. Laws and ethical standards differ between countries, and this question is beyond the scope of this report. Instead, we study fairness in a **technical and methodological sense**, following the definitions introduced in the course.

In this report, a model is considered *unfair* if it introduces **unjustified differences** in predictions or errors between groups. In other words, even when two individuals have similar medical risk, the model should not systematically make larger mistakes for one group than for another.

An important nuance is the following:

- **Different predicted outcomes can be justified** when groups have genuinely different underlying risks. In this dataset, smokers may incur higher medical costs, and reflecting this difference does not necessarily imply unfairness.

- **Unfairness becomes a concern** when the model systematically overprices or underprices one group, or when prediction errors are much larger for one group than for another, without a clear justification in the data.

For this reason, we focus on fairness criteria that evaluate **prediction errors and mispricing patterns**, rather than forcing equal predicted outcomes across groups. This distinction is important in insurance settings, where risk levels are not expected to be equal. As discussed in the fairness literature, Demographic Parity can be misleading when base rates differ between groups, whereas criteria such as Equal Opportunity and Equalized Odds provide more informative measures by comparing error rates across groups [5, 4, 1].

2 Data and problem formulation

2.1 Data source and variables

We use a synthetic but realistic U.S. health insurance dataset publicly available on Kaggle, originally published as *Insurance Data for Machine Learning* [7]. The dataset is synthetically generated but designed to reflect realistic relationships between demographic, lifestyle, and health-related variables and insurance charges[6]. All analyses and conclusions in this report are conditional on this dataset version.
The target variable is

$$y = \texttt{charges} \quad \text{(annual charges; continuous).}$$

The predictors include:

- Demographics: age, gender, region, number of children
- Lifestyle: BMI, exercise frequency, smoker status
- Health history: medical history, family medical history
- Insurance plan: coverage level
- Context: occupation

Dataset size. In our notebook run, the train–test split produced:

$$\text{Train shape } (750000, 22), \quad \text{Test shape } (250000, 22),$$

after one-hot encoding (22 predictors). This implies that 1'000'000 rows were used in this run.

2.2 Machine learning task

We consider a supervised regression problem, where the objective is to learn a function

$$\hat{y} = f_\theta(X),$$

mapping encoded predictors X to predicted annual insurance charges $\hat{y}$. Model parameters θ are estimated by minimizing standard regression loss functions (e.g., squared loss and regularized variants) on the training data. Model performance is then assessed on a held-out test set to evaluate generalization to unseen individuals.

2.3 Sensitive attribute

We treat smoking status as a sensitive attribute:

$$A \in \{0, 1\}, \quad A = 1 \text{ for smokers (yes).}$$

We do *not* remove smoking from the features because part of our question is whether the model reflects real differences without adding unfair errors.

3 Conceptual simulator and causal assumptions

A central idea in this course is that an observed dataset can be viewed as a realization of an underlying data-generating process. While this process cannot be fully identified from observational data alone, making causal assumptions explicit is essential for interpreting both predictive performance and fairness results[11, 9].

In this project, we therefore introduce a conceptual causal diagram (Directed Acyclic Graph, DAG) to formalize plausible relationships between the variables and insurance charges. The DAG is not used for causal estimation, but rather as a conceptual tool to guide fairness interpretation.

3.1 Causal diagram (DAG)

Figure 1 presents a high-level DAG representing assumed causal mechanisms underlying insurance charges, based on domain knowledge and the structure of the dataset.

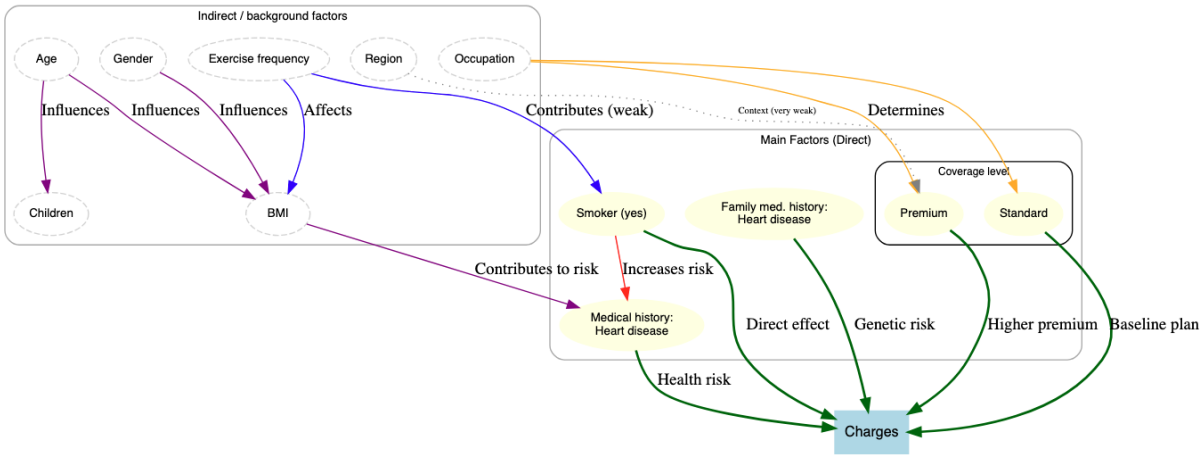


Figure 1: Conceptual DAG for insurance charges. Arrows indicate assumed causal influences.

The diagram distinguishes between three broad categories of variables:

Direct factors. Smoking status, medical history, family medical history, and insurance coverage level are assumed to have direct effects on insurance charges. These variables represent primary drivers of medical risk and pricing decisions.

Indirect or background factors. Variables such as age, gender, exercise frequency, body mass index (BMI), occupation, and region are assumed to influence charges indirectly. For example, age and exercise affect BMI, which in turn contributes to medical history, while occupation and region may influence insurance coverage choices.

Contextual factors. Some variables, notably region, are assumed to have weak or contextual effects. If a predictive model were to amplify such variables disproportionately, this could raise fairness concerns.

4 Pipeline and reproducibility

This section describes each step of our pipeline and the exact decisions made.

4.1 Step 1: Data loading and standardization of column names

We load `insurance.csv` and standardize column names by:

- trimming spaces,
- replacing spaces by underscores,
- converting to lowercase.

This prevents bugs caused by inconsistent naming.

4.2 Step 2: Type enforcement

We convert the variables `age`, `bmi`, `children`, and `charges` to numeric format, with invalid entries treated as missing values. All remaining variables (gender, smoker, region, medical history, family medical history, exercise frequency, occupation, and coverage level) are treated as categorical variables.

4.3 Step 3: Handling missing values

Two medical history columns contain missing values:

- `medical_history`
- `family_medical_history`

We replace missing entries with the explicit category *"No Record"*. Rationale: in medical datasets, missingness often means no recorded condition, not "unknown". Using a meaningful category avoids mixing missingness with real conditions and keeps group comparisons consistent.

Any missing numeric values introduced during numeric conversion are replaced with the median of the corresponding variable.

4.4 Step 4: One-hot encoding

We apply one-hot encoding to all categorical predictors (excluding target `charges`), with `drop_first=True` to avoid perfect multicollinearity. After encoding, the final feature matrix has 22 predictors in our run.

4.5 Step 5: Train–test split

We split the data into 75% training and 25% test:

$$(X_{\text{train}}, y_{\text{train}}),\ (X_{\text{test}}, y_{\text{test}}),$$

using a fixed random seed (`random_state=42`) to make results reproducible.

4.6 Step 6: Model training and selection

We train five regression models (details in Section 6). Models that are sensitive to scaling are trained in a pipeline with `StandardScaler`. We evaluate on the held-out test set using R^2, MAE, and RMSE and select the best model by RMSE.

4.7 Step 7: Fairness evaluation

We evaluate fairness for smokers vs non-smokers using:

- regression fairness (residuals, MAE, mispricing rates),
- bootstrap confidence intervals for group differences,
- classification-style fairness metrics (DP, EO, EOdds) after converting charges to a binary high-cost label,
- a stress test with a known injected bias (+5000 $ for smokers).

4.8 Reproducibility checklist and details

All experiments were conducted in Python using standard scientific libraries. To reproduce our results, a reader needs:

- the raw dataset file `insurance.csv` (same version as used in our experiments),
- the notebook or a script implementing the pipeline described above,
- a Python environment with the following libraries: numpy, pandas, scikit-learn, statsmodels, matplotlib, seaborn, and graphviz.

We split the data into training and test sets using a fixed random seed to ensure reproducibility:

$$(X_{\text{train}}, X_{\text{test}}, y_{\text{train}}, y_{\text{test}}) = \texttt{train_test_split}(X, y, \text{test_size} = 0.25, \text{random_state} = 42).$$

For models involving stochastic optimization or random subsampling, we fixed a global seed (`RANDOM_STATE`) to ensure reproducibility.

The following modeling choices were used:

- **Linear Regression**: trained in a pipeline with `StandardScaler`.
- **Elastic Net**: implemented via `ElasticNetCV` with l_1 ratios $\{0.1, 0.5, 0.9\}$, 25 logarithmically spaced α values in $[10^{-3}, 10^{2}]$, 5-fold cross-validation, and `max_iter=10000`.
- **HistGradientBoosting Regressor**: trained with default hyperparameters and a fixed random seed.
- **MLP Regressor**: trained in a pipeline with `StandardScaler`, two hidden layers of sizes (64, 32), ReLU activation, early stopping enabled, and `max_iter=500`.
- **GLM (Gamma with log link)**: fitted using `statsmodels` on unscaled data with an explicit intercept term.

The performance of the model was evaluated on the held test set using R^2, MAE, and RMSE. The model with the lowest test-set RMSE was selected as the baseline for fairness analysis.

5 Exploratory analysis: verifying group differences

Before training models, we verify whether the dataset itself shows a smoking-related cost difference.

5.1 Correlation with charges

Figure 2 shows correlations between encoded predictors and charges. Smoking status shows the strongest positive correlation with insurance charges, followed by premium coverage level and medical history indicators. Lifestyle and demographic variables such as BMI and age exhibit moderate influence, while region and occupation contribute minimally. This analysis confirms that smoking is a genuine cost driving factor present in the data itself, rather than an artifact of modeling. The correlation analysis highlights smoking as a key cost driver, but the fairness assessment requires understanding group-level cost distributions. We therefore compare the full distribution of insurance charges for smokers and non-smokers.

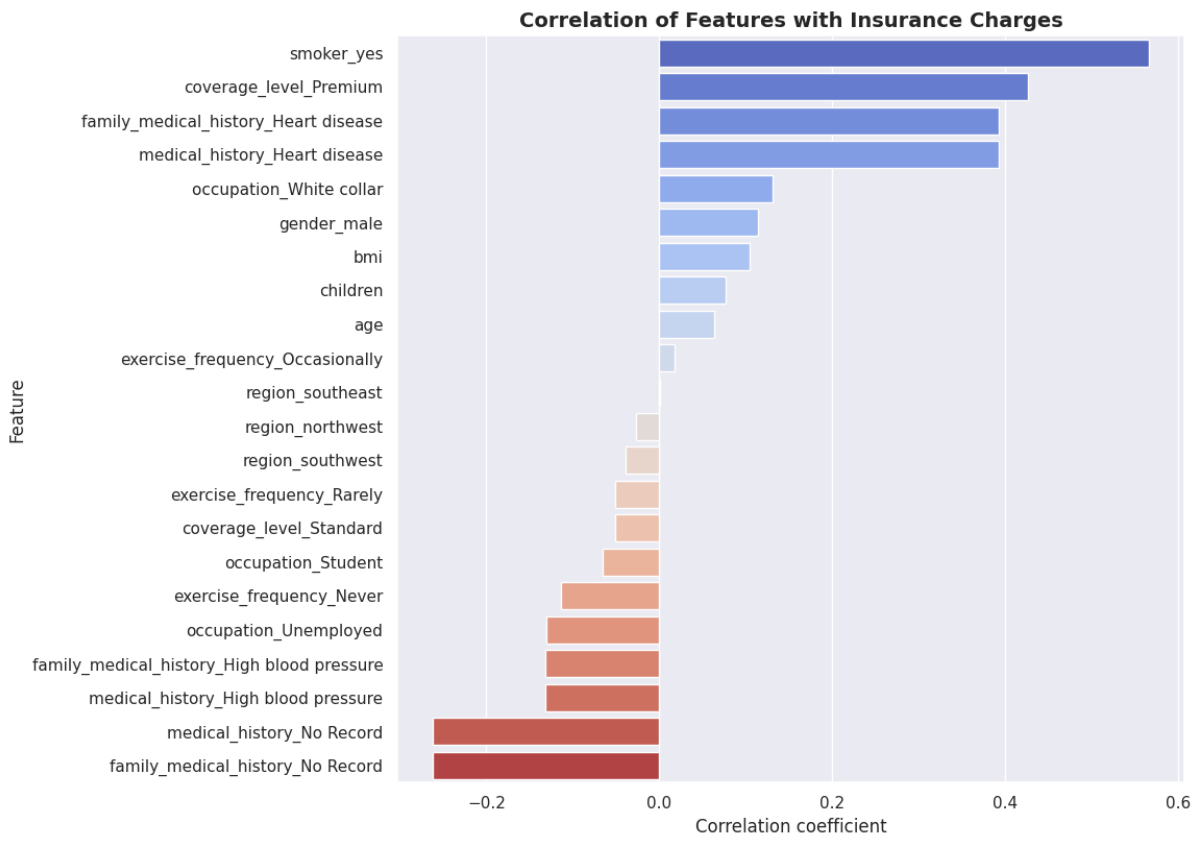


Figure 2: Correlation of features with insurance charges (from encoded dataset).

5.2 Distribution shift by smoking status

Figure 3 compares charges for smokers vs non-smokers. Smokers exhibit a distribution shifted to the right, with greater variance and a heavier upper tail. This indicates that smokers not only incur higher average costs but also face a higher likelihood of extreme medical expenses. These patterns establish the true underlying cost differences that any fair predictive model should reflect.

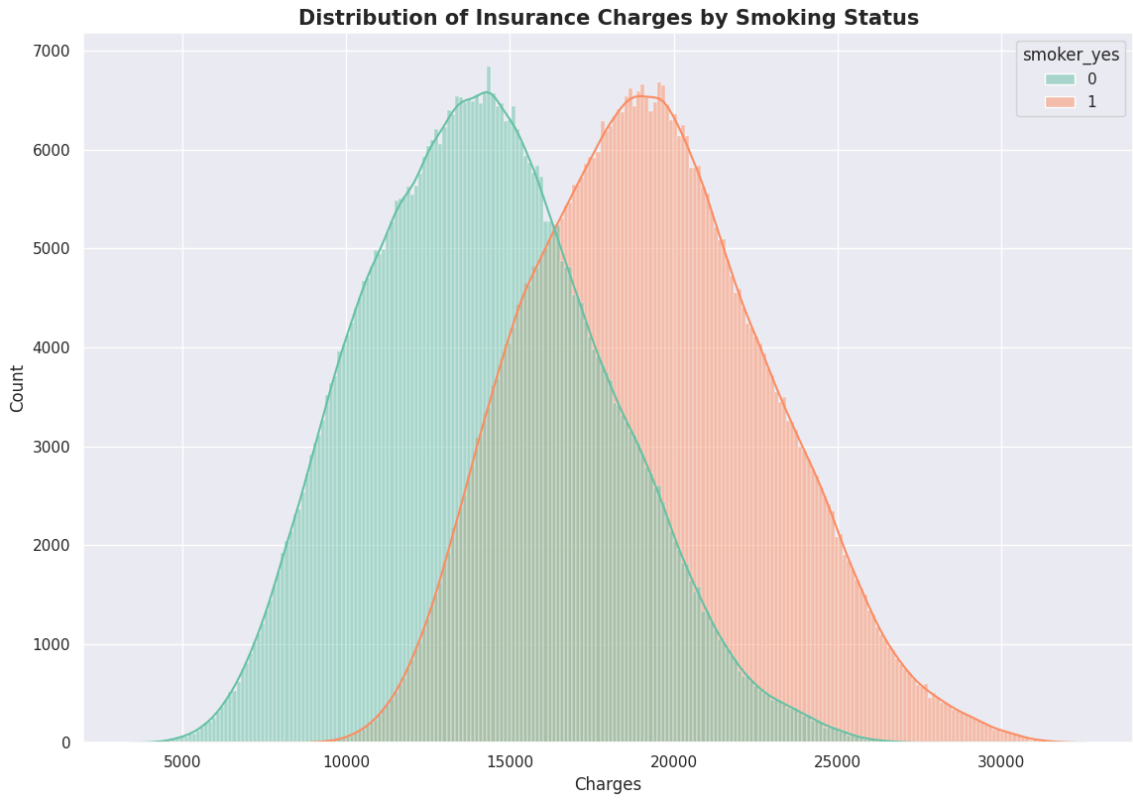


Figure 3: Distribution of annual insurance charges by smoking status.

Interpretation. These distributions highlight an important fairness principle: higher predicted costs for smokers are not, by themselves, evidence of unfairness, provided that smokers genuinely incur higher medical costs in the underlying data.

6 Predictive modeling

6.1 Models evaluated

We train and compare:

1. **Linear Regression (OLS).** Minimizes squared error; strong baseline and interpretable.
2. **Elastic Net (with CV).** Adds mixed L1/L2 regularization; hyperparameters selected by cross-validation.
3. **HistGradientBoosting Regressor.** Non-linear tree boosting method.
4. **MLP Regressor.** Neural network with hidden layers (64, 32), ReLU, early stopping.
5. **GLM (Gamma with log link).** A distributional model for positive costs, but may mismatch the data.

6.2 Evaluation metrics

On the test set we report:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \quad \text{MAE} = \frac{1}{n}\sum_i |y_i - \hat{y}_i|, \quad \text{RMSE} = \sqrt{\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2}.$$

RMSE is particularly relevant in insurance pricing, as it places greater weight on large errors that can lead to substantial financial mispricing.

6.3 Test-set results and selection

Table 1 reports test-set performance (numbers from our notebook run).

Table 1: Predictive performance on the test set.

Model	R^2	MAE ($)	RMSE ($)
Linear Regression	0.996	250.291	288.892
Elastic Net CV	0.996	250.293	288.894
MLP Regressor	0.995	256.421	299.070
HistGradientBoosting	0.994	274.846	328.564
GLM (Gamma Log)	0.957	622.795	911.947

Conclusion (model choice). We select **Linear Regression** as the baseline fairness model because it is best in accuracy, stable, fast, and easy to interpret and audit.

6.4 Cross-validation performance

Test-set performance depends on a single train–test split and may be sensitive to sampling variability. To verify that the relative performance of the models is stable, we additionally evaluate them using 3-fold cross-validation on the full dataset.

Cross-validation provides a more reliable estimate of generalization performance by averaging results across multiple data splits. This is particularly relevant for insurance data, where cost distributions are skewed and a small number of high-cost individuals can disproportionately affect performance estimates.

Table 2 reports the mean R^2 score across folds for each model.

Table 2: Cross-validation performance (mean R^2 across 3 folds).

Model	Mean R^2
Linear Regression	0.9957
Elastic Net (CV)	0.9957
MLP Regressor	0.9954
HistGradientBoosting	0.9945

Interpretation. The cross-validation results confirm that model performance is stable across different data splits. Linear Regression and Elastic Net achieve the highest mean R^2 and perform nearly identically across folds. This indicates that the strong test-set performance of the linear model is not an artifact of a particular split, but reflects robust generalization.

6.5 Diagnostics: predicted vs actual

Figure 4 displays the predicted insurance charges against the true observed charges for the selected model. Each point represents one individual, and the red diagonal line corresponds to the perfect prediction ($\hat{y} = y$).

The scatter plot shows that the predicted charges align closely with the actual values across the entire range of costs. Most points lie tightly along the diagonal, indicating **high predictive accuracy** and the absence of systematic overprediction or underprediction. Importantly, the model remains stable even for people with very high medical charges, where prediction errors could otherwise be amplified.

This strong alignment suggests that the model successfully captures the underlying cost structure of the dataset and provides a reliable foundation for the fairness analysis presented in the following section.

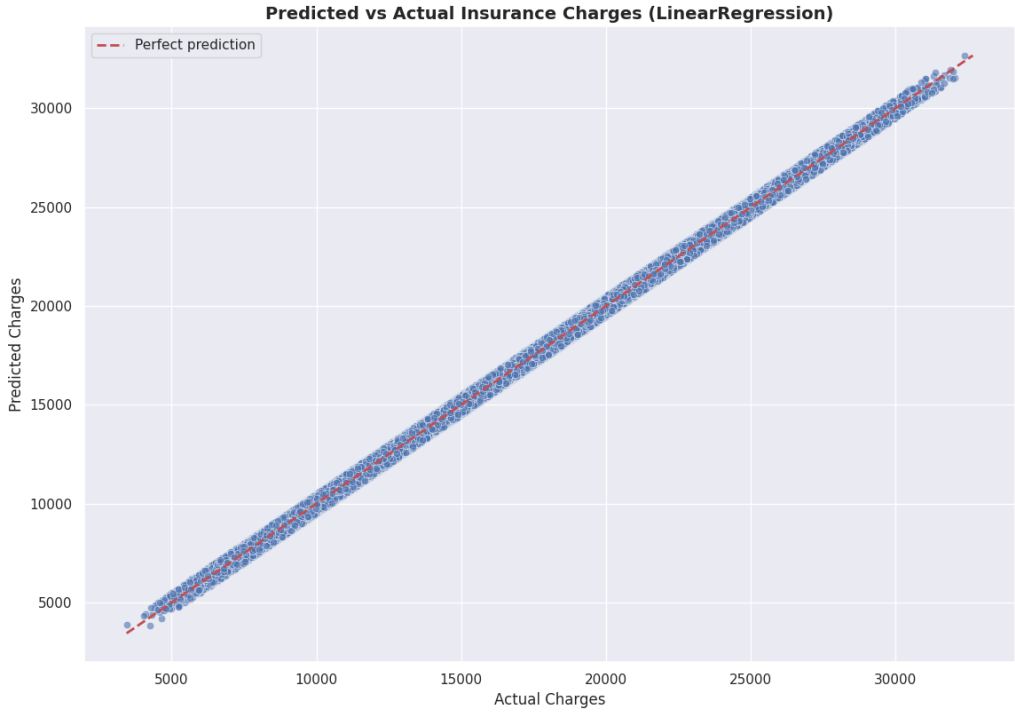


Figure 4: Predicted vs actual charges for the selected model.

6.6 Interpretability: influential features (linear model)

To better understand the behavior of the selected linear model, we examine the magnitude of its estimated coefficients. Because we standardize all predictors (including one-hot indicators), coefficient magnitudes reflect the association strength per standard-deviation change of each feature, making them comparable as a relative influence diagnostic.

Figure 5 presents the ten predictors with the largest absolute coefficient magnitudes. Smoking status and insurance coverage level emerge as the most influential features, which is consistent with both domain knowledge and the exploratory analysis. Health- and lifestyle-related variables, such as body mass index, also contribute substantially to the predictions.

Importantly, the strong effect of smoking status does not automatically imply unfairness. Instead, it reflects real differences in expected medical costs present in the data. This highlights the importance of evaluating fairness through prediction errors and pricing behavior, rather than through the mere use of group-related variables.

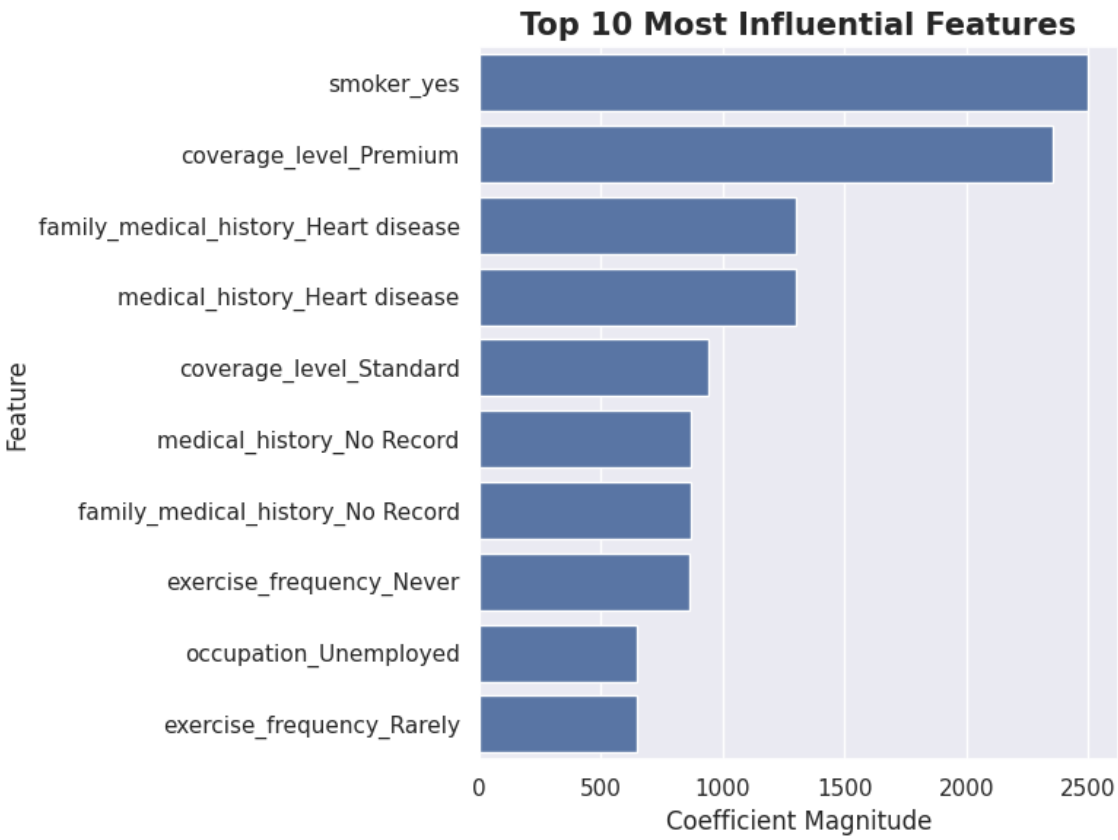


Figure 5: Top 10 influential features (absolute coefficient magnitude).

Having established strong predictive performance, we now turn to the central question of whether the selected model behaves fairly across smoking groups.

7 Fairness analysis (smokers vs non-smokers)

We measure fairness using complementary perspectives.

7.1 Notation

Let $A \in \{0, 1\}$ be smoking status (1=smoker). For each test individual i:

$$r_i = y_i - \hat{y}_i \quad \text{(residual)}.$$

Positive residuals indicate underprediction; negative residuals indicate overprediction.

7.2 Regression fairness: residuals and mispricing rates

We compute group summaries:

- mean residual (bias),
- MAE (average error magnitude),
- underpricing rate: $\hat{y} < 0.9y$,
- overpricing rate: $\hat{y} > 1.1y$.

This residual-based evaluation can be interpreted as a group-conditional calibration check: if a model is well-calibrated within each group, residuals should be centered near zero with comparable dispersion.

Table 3 reports baseline results (from our notebook run).

Table 3: Residual-based fairness summary by smoking status (baseline model).

Group	Count	Mean residual	MAE	Underpricing rate	Overpricing rate
Non-smokers (no)	124949	0.370625	250.448153	0.000008	0.000040
Smokers (yes)	125051	0.871519	250.133503	0.000000	0.000000

Interpretation. The results indicate that the model behaves very similarly for smokers and non-smokers. The mean residuals are close to zero in both groups, showing no systematic tendency to overpredict or underpredict charges for either group. The MAE values are nearly identical (around 250 $), which implies that the average prediction error has the same magnitude for smokers and non-smokers.

In addition, the underpricing and overpricing rates are extremely small and do not differ significantly between groups. Together, these findings suggest that the model does not introduce systematic pricing errors based on smoking status. Instead, it achieves comparable accuracy and error behavior for both groups.

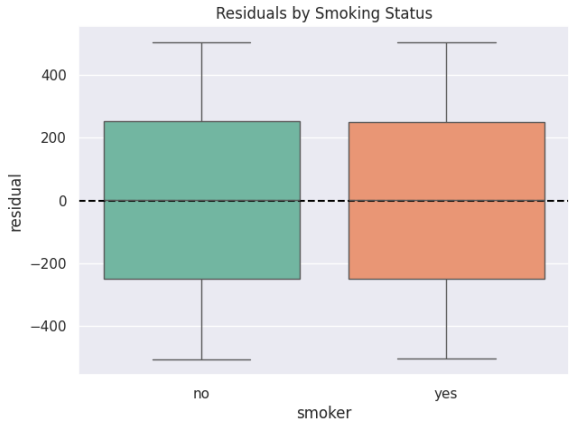
7.3 Bootstrap inference for group differences

To test whether group differences are meaningful or due to noise, we compute 95% bootstrap confidence intervals (2000 resamples) for the smoker minus non-smoker difference in key metrics.

(a) Bootstrap 95% confidence intervals for group differences (Smokers − Non-smokers).

Metric (difference)	2.5%	97.5%
Mean residual	-1.710	2.914
MAE	-1.486	0.800
Underpricing rate	-0.000	0.000
Overpricing rate	-0.000	0.000

(b) Distribution of prediction residuals by smoking status. The dashed line at zero indicates perfect prediction.



Interpretation. As shown in Table 4a, all bootstrap confidence intervals include zero, indicating that the observed differences in error metrics between smokers and non-smokers are not statistically significant. This suggests that the small numerical differences reported in the point estimates are compatible with random sampling variation rather than systematic group-level bias.

In particular, the confidence intervals for the mean residual and MAE are narrow and centered close to zero, reinforcing the conclusion that both groups experience comparable prediction accuracy. Similarly, the intervals for underpricing and overpricing rates collapse to zero, confirming that extreme mispricing events are equally rare for smokers and non-smokers.

Consistent with these numerical results, the residual distributions shown in Figure 4b are centered around zero and show a similar spread between groups. Taken together, the bootstrap confidence intervals and the residual distribution provide strong statistical evidence that the model does not introduce differential prediction errors by smoking status. This strengthens the conclusion that the baseline model satisfies the regression-based fairness criteria under uncertainty.

7.4 Classification-style fairness metrics (Lab 7 adaptation)

To apply the fairness metrics introduced in Lab 7, which are defined for binary classification tasks, we convert the continuous insurance charges into a binary outcome:

$$Y^{\text{high}} = \begin{cases} 1, & \text{if } y > q_{0.75}, \\ 0, & \text{otherwise}, \end{cases}$$

where $q_{0.75}$ denotes the 75th percentile of the charge distribution in the test set. The predicted charges are binarized using the same threshold.

We then compute the following group fairness metrics:

- Demographic Parity Difference (DP),
- Equal Opportunity Difference (EO),
- Equalized Odds Difference (EOdds),

defined as absolute differences between smokers and non-smokers. Smaller values indicate more similar behavior between groups[5, 13].

Table 5: Group fairness metrics (baseline model) after binarizing charges into high-cost vs low-cost.

Metric	Value
Demographic Parity Difference (DP)	0.3535
Equal Opportunity Difference (EO)	0.0340
Equalized Odds Difference (EOdds)	0.0512

Interpretation. The Demographic Parity (DP) difference is relatively large because smokers are genuinely more likely to belong to the high-cost group in the data. Consequently, DP mainly reflects the underlying differences in the prevalence of outcomes rather than the unfair behavior introduced by the model, making it a weak indicator of unfairness in this setting.

In contrast, the Equal Opportunity (EO) and Equalized Odds (EOdds) differences are small. This indicates that the model achieves similar true-positive rates and comparable error patterns for smokers and non-smokers, and it suggests that its predictive performance is balanced across groups.

7.5 Stress test: injecting known discrimination (+5000 $ for smokers)

Let $A \in \{0, 1\}$ denote smoking status, where $A = 1$ corresponds to smokers and $A = 0$ to non-smokers. We validate the fairness pipeline by creating an unfair prediction rule:

$$\hat{y}^{\text{biased}} = \begin{cases} \hat{y} + 5000, & \text{if } A = 1, \\ \hat{y}, & \text{if } A = 0. \end{cases}$$

This intervention deliberately violates fairness by adding a constant surcharge that is not related to individual risk. We recompute the fairness metrics.

Table 6: Residual-based fairness summary by smoking status (artificially biased model: +5000 $ for smokers).

Group	Count	Mean residual	MAE	Underpricing rate	Overpricing rate
Non-smokers (no)	124949	0.370625	250.448153	0.000008	0.000040
Smokers (yes)	125051	-4999.128481	4999.128481	0.000000	1.000000

Table 7: Fairness metrics comparison: baseline vs biased model (+5000 $ for smokers).

Model	DP	EO	EOdds
Baseline model	0.3535	0.0340	0.0512
Biased model (+5000 $)	0.8150	0.0662	0.8648

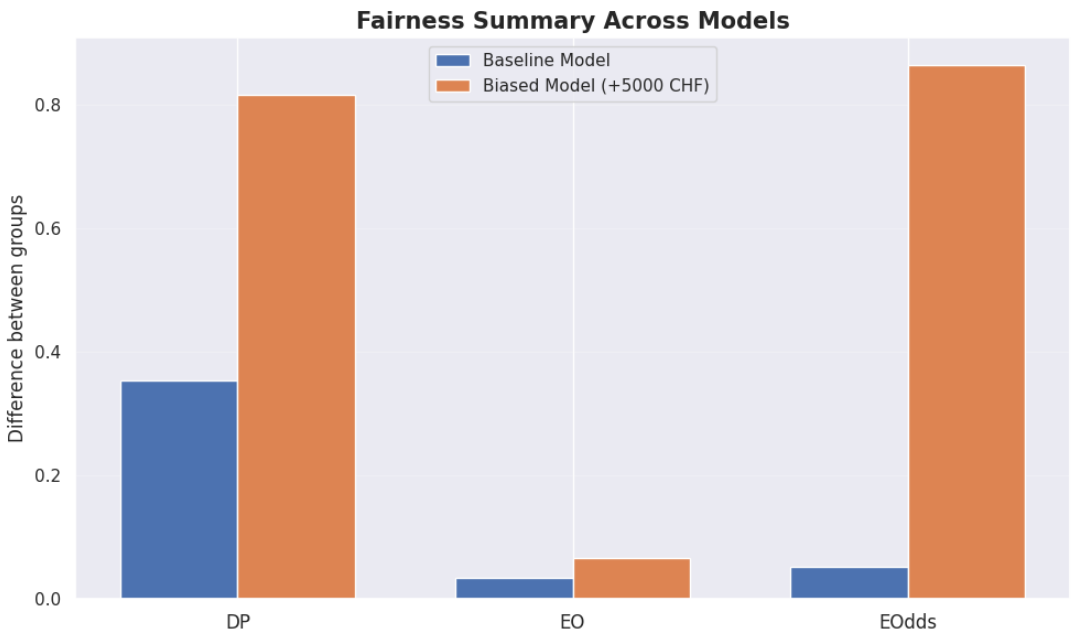


Figure 6: Fairness dashboard comparing baseline vs biased predictions.

Interpretation. Table 6 illustrates the effect of the artificially biased prediction rule on regression-based fairness metrics. For non-smokers, residual statistics remain essentially unchanged compared to the baseline model. In contrast, smokers exhibit a dramatic shift in prediction errors: the mean residual is close to -5000 \$, the MAE increases to nearly 5000 \$, and the overpricing rate reaches 1.0. This indicates that every smoker is systematically overcharged by the same amount, which confirms the presence of severe and intentional discrimination.

Table 7 further highlights this disparity through classification-style fairness metrics. Although the baseline model shows relatively small differences in Equal Opportunity (EO) and Equalized Odds (EOdds), the biased model produces a substantial increase in all metrics, most notably in Equalized Odds. This reflects large discrepancies in both true-positive and false-positive rates between smokers and non-smokers, consistent with a strongly discriminatory decision rule.

These effects are visually summarized in Figure 6, which contrasts the baseline and biased models on multiple fairness indicators. The biased model exhibits pronounced deviations in all metrics, whereas the baseline model remains comparatively balanced. Together, the regression-based metrics, classification-style fairness measures, and visual diagnostics provide consistent evidence that the fairness evaluation pipeline reliably detects discrimination when it is present. This stress test therefore validates that the earlier fairness conclusions for the baseline model are meaningful and not an artifact of insensitive metrics.

8 Discussion

8.1 Answer to Q1: Do smokers have higher costs in the data?

Exploratory analysis clearly shows that smokers incur higher annual insurance charges and exhibit more extreme cost values than non-smokers. This pattern is consistent in summary statistics, correlation analysis, and distributional plots. These findings align with the domain knowledge in health economics and support the interpretation that smoking is a genuine risk factor in the dataset rather than an artifact of the modeling process.

8.2 Answer to Q2: Does the model introduce extra unfairness?

Across both the regression fairness checks and the Lab 7 (classification-style) checks, we find no sign that the model makes systematically larger or different errors for smokers than for non-smokers. The main result is that the model's error patterns look similar between the two groups, although smokers have higher charges in the data itself (Section 7).

8.3 What fairness does and does not mean here

From an ethical and business perspective, fairness in insurance pricing does not imply equal predicted costs between groups. Rather, it requires alignment with true medical risk, the absence of arbitrary penalties, and comparable predictive accuracy between individuals. Under these criteria, the baseline model satisfies all three conditions, while the artificially biased model clearly violates them and is correctly identified by the fairness diagnostics.

The conclusions of this analysis should be interpreted in a technical sense. In this context, fairness refers to comparable predictive accuracy and similar error patterns between groups, rather than identical predicted outcomes. Importantly, this study does not claim that the use of smoking status in insurance pricing is ethically or legally acceptable in all settings. If regulatory or policy constraints require limiting or excluding the use of smoking status, a different modeling objective and evaluation framework would be necessary.

8.4 Limitations

This study has several limitations. First, fairness is evaluated with respect to a single sensitive attribute, smoking status, while potential intersectional effects involving combinations of attributes such as gender, region, or occupation are not considered. Second, the dataset is synthetic and the observed relationships may differ from those found in real-world insurance claims data. Third, some fairness metrics depend on the chosen threshold for defining high-cost individuals, here set at the 75th percentile, which introduces an element of arbitrariness. Finally, the analysis focuses on auditing the fairness of standard predictive models and does not explore fairness-aware training methods or constrained optimization approaches.

8.5 Future work

Several extensions could deepen and refine the fairness analysis presented in this report. First, while smoking status was examined as a single sensitive attribute, future work could investigate *intersectional fairness*, for example by jointly considering smoking status with age, gender, or coverage level, to identify whether fairness conclusions remain stable across more granular subgroups.

Second, the present analysis evaluates fairness *after* model training. An important extension would be to compare the baseline approach with *fairness-aware learning strategies*, such as regularization-based constraints or post-processing methods that explicitly control error disparities, and to study the resulting trade-offs between predictive accuracy and fairness[9].

Third, although the causal diagram was used for interpretation, future work could move toward a more explicit *causal fairness* framework. This would involve assessing how fairness conclusions change under plausible counterfactual interventions suggested by the DAG, such as modifying lifestyle-related variables while holding other factors fixed.

Finally, applying the proposed fairness pipeline to real-world insurance claims data would be a valuable next step, as real datasets may exhibit stronger noise, measurement error, and institutional biases that are not fully captured in synthetic data.

9 Conclusion

This project examined the interaction between predictive accuracy and fairness in health insurance pricing, with smoking status treated as a sensitive attribute. Using a realistic insurance dataset, we first established that smokers incur substantially higher medical charges than non-smokers, reflecting genuine differences in underlying health risk rather than modeling artifacts. Among the models evaluated, linear regression emerged as a strong and stable baseline and was therefore selected for the fairness audit.

Fairness was evaluated through multiple complementary lenses, including residual-based regression metrics, bootstrap inference for group differences, and classification-style fairness measures adapted from the course. Across all these perspectives, the baseline model exhibited comparable error magnitudes, similar residual distributions, and nearly identical mispricing rates for smokers and non-smokers. These results indicate that the model reproduces true cost differences without introducing systematic overpricing, underpricing, or unequal prediction errors across groups.

To assess whether these conclusions were meaningful rather than an artifact of insensitive metrics, we conducted a controlled stress test by introducing an artificial surcharge for smokers. This intervention produced large and consistent disparities across all fairness indicators, particularly in error-based and outcome-conditional metrics. The stress test therefore confirms that the fairness evaluation pipeline is capable of detecting discrimination when it is present and validates the reliability of the conclusions drawn for the baseline model.

This study shows how fairness evaluation can distinguish between differences that are justified by underlying risk and disparities introduced by predictive models themselves in high-stakes decision-making contexts.

Final statement

Although smokers exhibit higher true medical costs in the data, the selected predictive model captures these differences without introducing additional unfair bias under the fairness definitions considered in this report. The results highlight that fairness in insurance pricing does not require equal predicted outcomes across groups, but rather comparable predictive accuracy and the absence of unjustified error disparities. This distinction is essential for the responsible deployment of machine learning models in socially and economically consequential settings such as insurance pricing.

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A Appendix: Metric definitions and thresholds

A.1 Mispricing rates

Underpricing: $\hat{y} < 0.9y$.
Overpricing: $\hat{y} > 1.1y$.

A.2 High-cost threshold

High-cost label defined by test-set 75th percentile:

$$Y^{\text{high}} = \begin{cases} 1, & \text{if } y > q_{0.75}, \\ 0, & \text{otherwise,} \end{cases}$$

where $q_{0.75}$ denotes the 75th percentile of the charge distribution in the test set.

A.3 Fairness metrics (Lab 7 adaptation)

Let $\hat{Y}$ be predicted high-cost and A be smoking status.

- DP difference: $|\mathbb{P}(\hat{Y} = 1|A = 0) - \mathbb{P}(\hat{Y} = 1|A = 1)|$
- EO difference: $|\mathbb{P}(\hat{Y} = 1|Y = 1, A = 0) - \mathbb{P}(\hat{Y} = 1|Y = 1, A = 1)|$
- EOdds difference: sum of TPR and FPR absolute gaps across groups.